

2390. Removing Stars From a String

Solved 

Medium

 Topics

 Companies

 Hint

You are given a string `s`, which contains stars `*`.

In one operation, you can:

- Choose a star in `s`.
- Remove the closest **non-star** character to its **left**, as well as remove the star itself.

Return *the string after all stars have been removed*.

Note:

- The input will be generated such that the operation is always possible.
- It can be shown that the resulting string will always be unique.

Example 1:

Input: `s = "leet**cod*e"`

Output: `"lecoe"`

Explanation: Performing the removals from left to right:

- The closest character to the 1st star is 't' in `"leet**cod*e"`. `s` becomes `"lee*cod*e"`.
- The closest character to the 2nd star is 'e' in `"lee*cod*e"`. `s` becomes `"lecod*e"`.
- The closest character to the 3rd star is 'd' in `"lecod*e"`. `s` becomes `"lecoe"`.

There are no more stars, so we return `"lecoe"`.

Example 2:

Input: `s = "erase*****"`

Output: `""`

Explanation: The entire string is removed, so we return an empty string.

Constraints:

- `1 <= s.length <= 105`
- `s` consists of lowercase English letters and stars `*`.
- The operation above can be performed on `s`.

```

// Time Complexity - O(N)
// Run Time - 39ms
class Solution {
    public String removeStars(String s) {
        Deque<Character> monotonicStack = new ArrayDeque<>();
        for (int i = 0; i < s.length(); i++) {
            if (s.charAt(i) != '*') {
                monotonicStack.addLast(s.charAt(i));
            } else {
                if (!monotonicStack.isEmpty()) {
                    monotonicStack.pollLast();
                }
            }
        }
        StringBuilder result = new StringBuilder();
        while (!monotonicStack.isEmpty()) {
            result.append(monotonicStack.pollFirst());
        }

        return result.toString();
    }
}

```

```

// Time Complexity - O(N)
// Run Time - 28ms
class Solution {
    public String removeStars(String s) {
        StringBuilder result = new StringBuilder();
        for (char c : s.toCharArray()) {
            if (c == '*') {
                result.deleteCharAt(result.length() - 1);
            } else {
                result.append(c);
            }
        }

        return result.toString();
    }
}

```